FAST- National University of Computer and Emerging Sciences

Theory of Automata

Assignment # 2

SUBMITTED TO: Maam Maryam Shahbaz

SUBMITTED BY: Muhammad Moiz Ansari

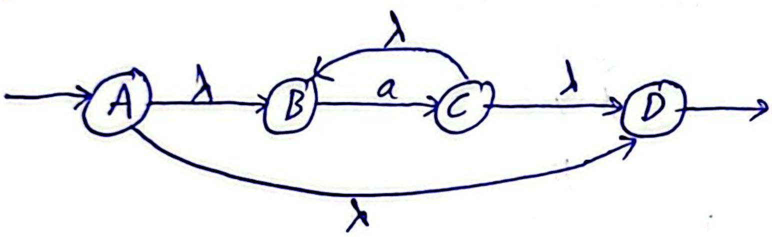
SECTION: BS(CS) - F

Roll No: 23i - 0523

Q1:

a. a^*

(a)



(b)

ε-NFA Table:

	a	b	λ
A	ϕ	ϕ	B, D
B	C	ϕ	ϕ
C	ϕ	ϕ	B, D
D	ϕ	ϕ	ϕ

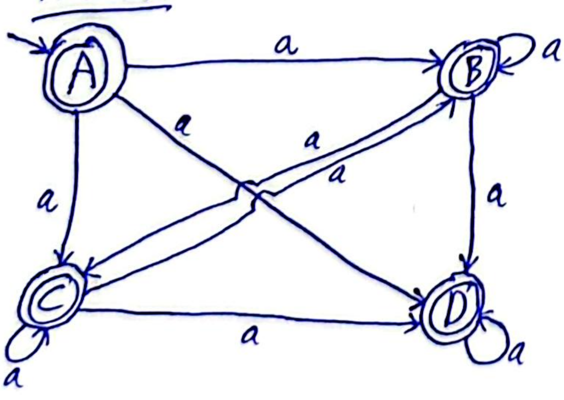
ε-Closures:

A: { A, B, D }
B: { B }
C: { B, C, D }
D: { D }

NFA Table:

	a	b
A	{ B, C, D }	ϕ
B	{ B, C, D }	ϕ
C	{ B, C, D }	ϕ
D	{ D }	ϕ

NFA:

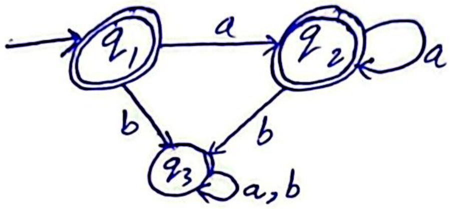


②

(c)

DFA:

	a	b
∓ $q_1 = A$	$BCD = q_2$	$\phi = q_3$
+ $q_2 = BCD$	$BCD = q_2$	$\phi = q_3$
$q_3 = \phi$	$\phi = q_3$	$\phi = q_3$

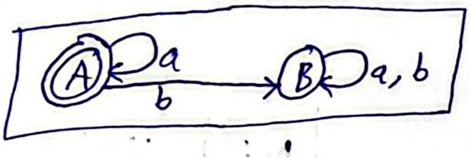


(d)

	a	b
−+ q_1	q_2	q_3
+ q_2	q_2	q_3
q_3	q_3	q_3

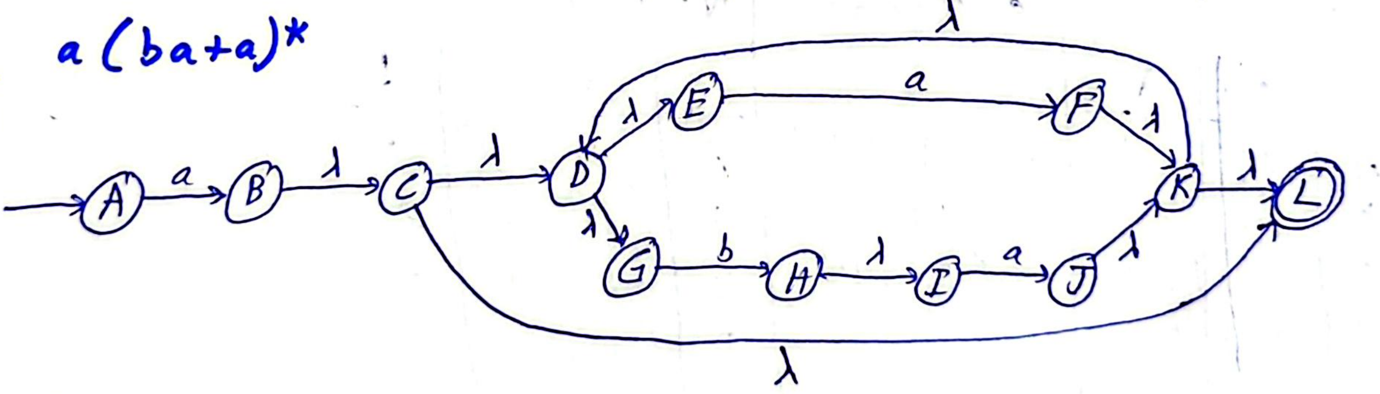
	Accepting	Non-accepting
	G1	G2
0-Eq:	$\{q_1, q_2\}$	$\{q_3\}$
	A	B

Minimized:

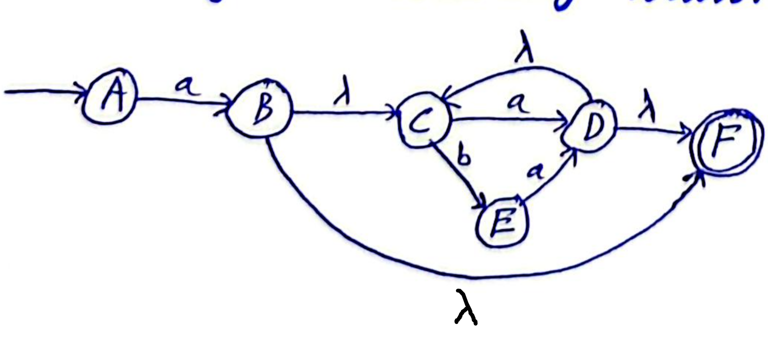


b. $a(ba+a)^*$

(a)



Removing unnecessary nulls:



(b)

ε-NFA Table:

	a	b	λ
A	B	ϕ	ϕ
B	ϕ	ϕ	C, F
C	D	E	ϕ
D	ϕ	ϕ	C, F
E	D	ϕ	ϕ
F	ϕ	ϕ	ϕ

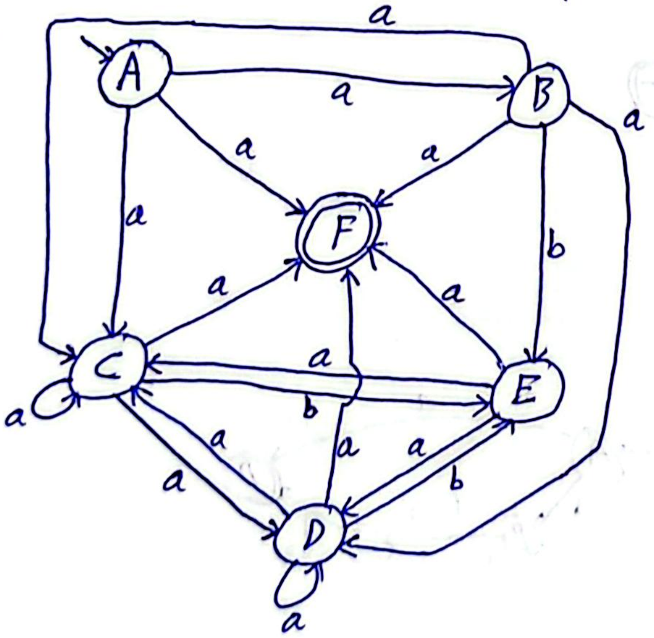
ε-Closures

A: {A}
B: {B, C, F}
C: {C}
D: {C, D, F}
E: {E}
F: {F}

NFA Table:

	a	b
A	{B, C, F}	ϕ
B	{C, D, F}	E
C	{C, D, F}	E
D	{C, D, F}	E
E	{C, D, F}	ϕ
F	ϕ	ϕ

NFA:

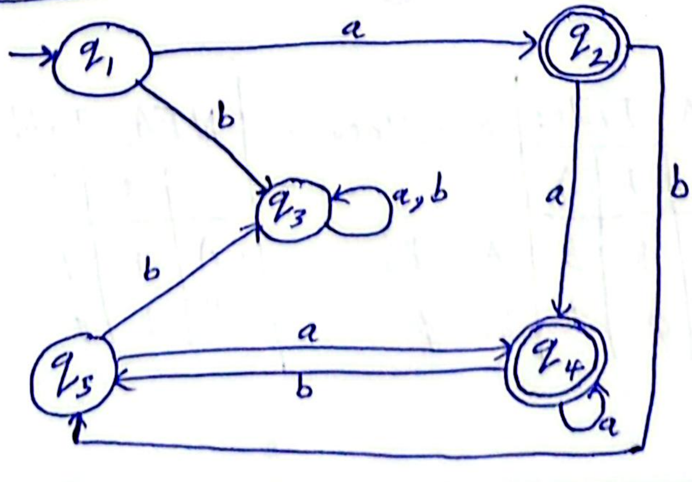


(c)

DFA Table:

	a	b
− A = q_1	BCF = q_2	ϕ = q_3
+ BCF = q_2	CDF = q_4	E = q_5
ϕ = q_3	ϕ = q_3	ϕ = q_3
+ CDF = q_4	CDF = q_4	E = q_5
E = q_5	CDF = q_4	ϕ = q_3

DFA:



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(d)

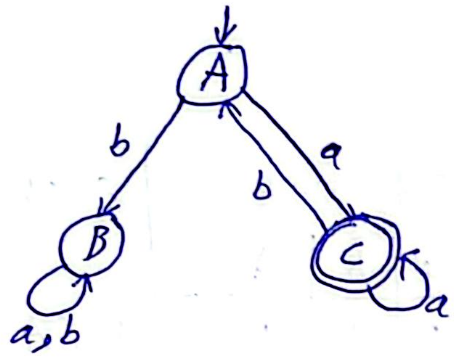
DFA Table:

	a	b
q_1	q_2	q_3
q_2	q_4	q_5
q_3	q_3	q_3
q_4	q_4	q_5
q_5	q_4	q_3

n-Equivalences:

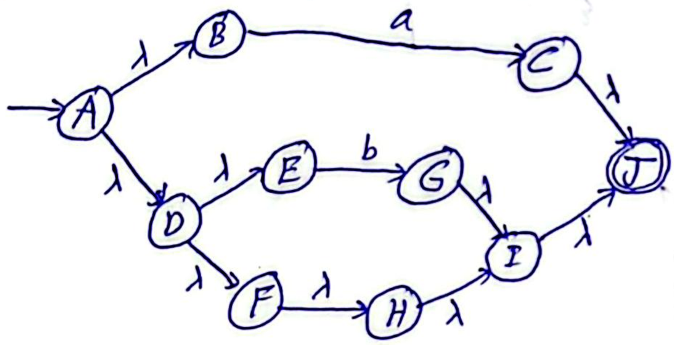
	G1	G2
0-Eq:	{1,3,5}	{2,4}
1-Eq:	{1,5}{3}	{2,4}
2-Eq:	{1,5}{3}	{2,4}
	A B	C

Minimized DFA:

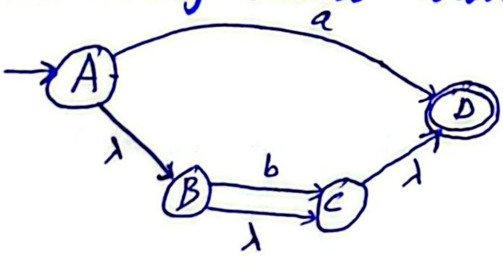


C. $a + b + \lambda$

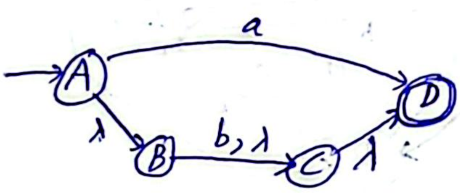
(a)



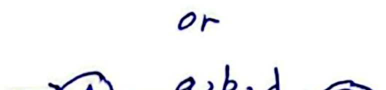
Removing excess nulls:



or



or



(b)

ε-NFA Table:

	a	b	λ
A	B	B	B
B	ϕ	ϕ	ϕ

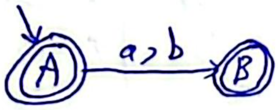
ε-Closures:

A: {B}
B: ϕ

NFA Tables:

	a	b
±A	B	B
+B	ϕ	ϕ

NFA:

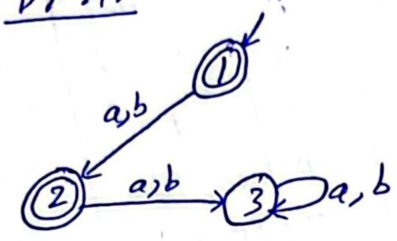


(c)

DFA Table:

	a	b
A=1	B=2	B=2
B=2	ϕ=3	ϕ=3
ϕ=3	ϕ=3	ϕ=3

DFA:



(d)

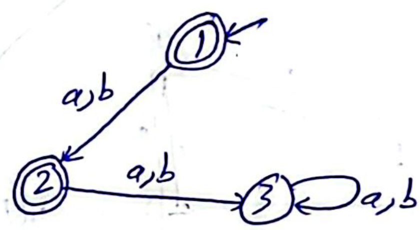
DFA Table:

	a	b
1	2	2
2	3	3
3	3	3

n-Equivalences:

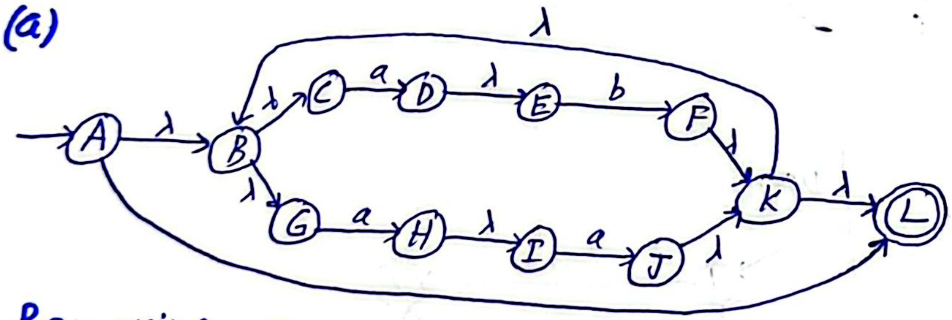
	G1	G2
0-Eq:	{1,3}	{2}
1-Eq:	{1} {3}	{2}

Minimized DFA:

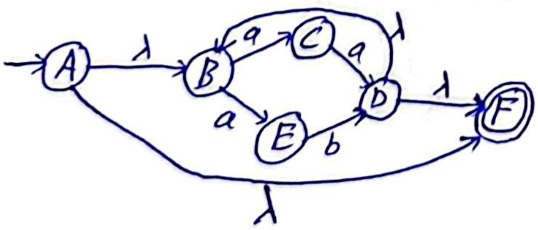


d. $(ab+aa)^*$

(a)



Removing excess nulls:



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(b)

ε-NFA Table:

	a	b	λ
A	ϕ	ϕ	B, F
B	C, E	ϕ	ϕ
C	ϕ	D	ϕ
D	ϕ	ϕ	B, F
E	D	ϕ	ϕ
F	ϕ	ϕ	ϕ

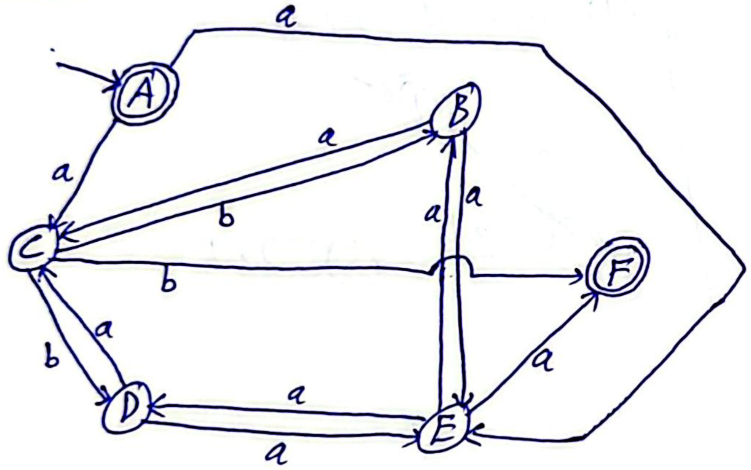
ε-Closures:

A: {A, B, F}
B: {B}
C: {C}
D: {B, D, F}
E: {E}
F: {F}

NFA Table:

	a	b
→ A	{C, E}	ϕ
B	{C, E}	ϕ
C	ϕ	{B, D, F}
D	{C, E}	ϕ
E	{B, D, F}	ϕ
+ F	ϕ	ϕ

NFA:

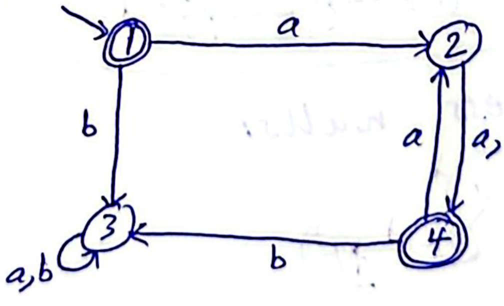


(c)

DFA Table:

	a	b
-+ A = 1	CE = 2	ϕ = 3
CE = 2	BDF = 4	BDF = 4
ϕ = 3	ϕ = 3	ϕ = 3
+ BDF = 4	CE = 2	ϕ = 3

DFA:



(d)

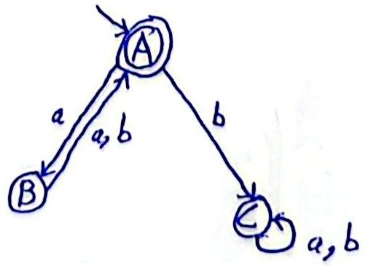
DFA Table:

	a	b
-+ 1	2	3
2	4	4
3	3	3
+ 4	2	3

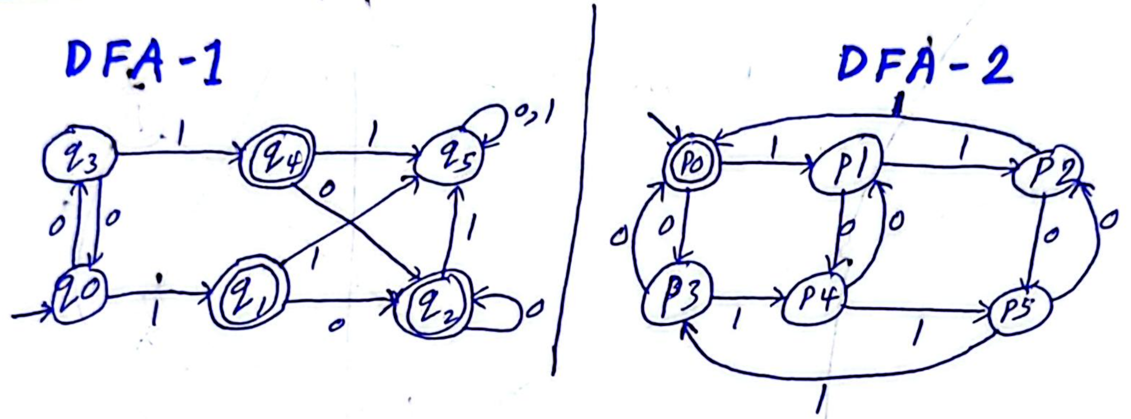
n-Equivalences:

	G1	G2
0-Eq:	{1, 4}	{2, 3}
1-Eq:	{1, 4}	{2} {3}

Minimized DFA:



Q2:

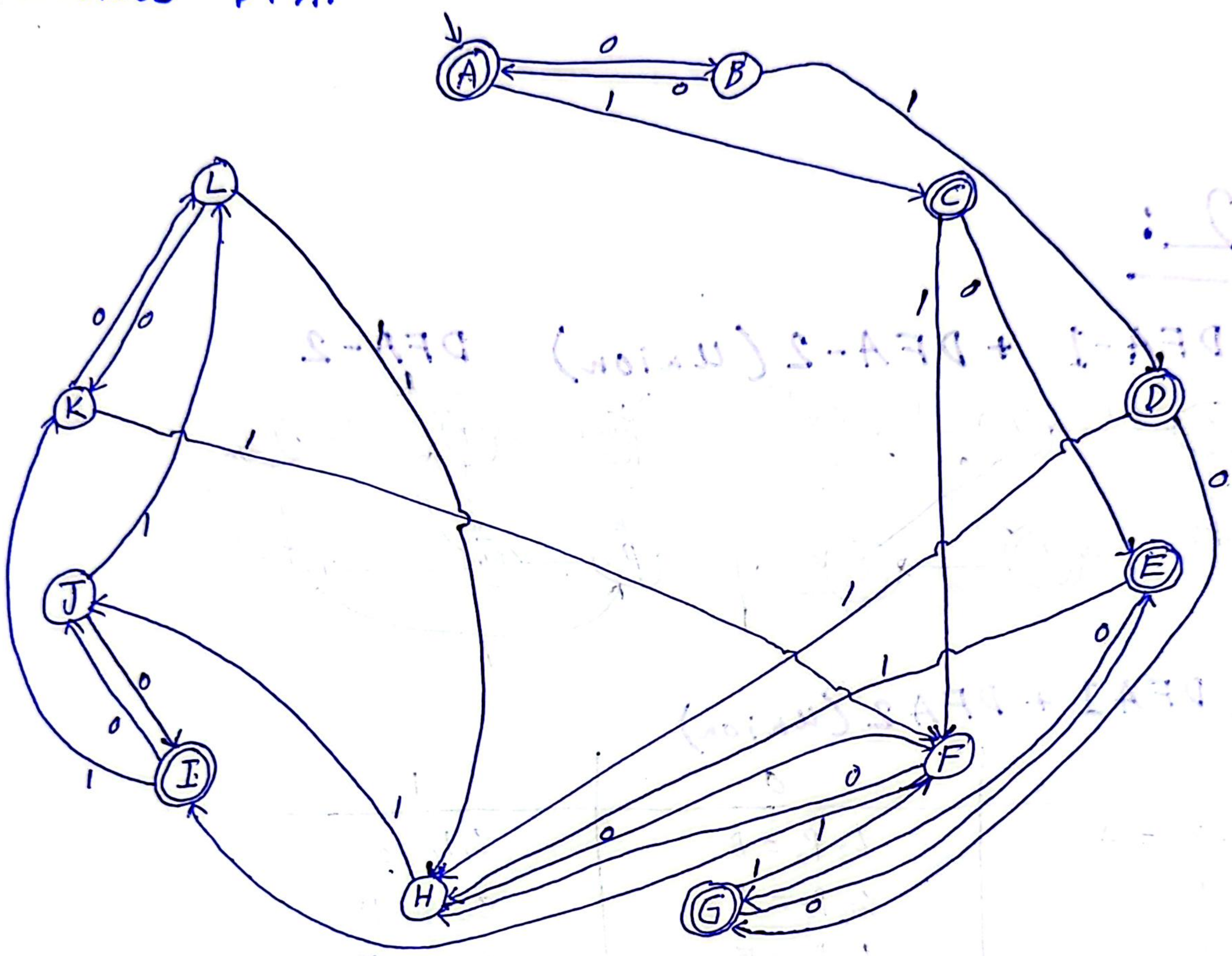


a. DFA1 + DFA2 (union)

States	0	1
−+ $q_0 P_0 = A$	$q_3 P_3 = B$	$q_1 P_1 = C$
$q_3 P_3 = B$	$q_0 P_0 = A$	$q_4 P_4 = D$
+ $q_1 P_1 = C$	$q_2 P_4 = E$	$q_5 P_2 = F$
+ $q_4 P_4 = D$	$q_2 P_1 = G$	$q_5 P_5 = H$
+ $q_2 P_4 = E$	$q_2 P_1 = G$	$q_5 P_5 = H$
$q_5 P_2 = F$	$q_5 P_5 = H$	$q_5 P_0 = I$
+ $q_2 P_1 = G$	$q_2 q_4 = E$	$q_5 P_2 = F$
$q_5 P_5 = H$	$q_5 P_2 = F$	$q_5 P_3 = J$
+ $q_5 P_0 = I$	$q_5 P_3 = J$	$q_5 P_1 = K$
$q_5 P_3 = J$	$q_5 P_0 = I$	$q_5 P_4 = L$
$q_5 P_1 = K$	$q_5 P_4 = L$	$q_5 P_2 = F$
$q_5 P_4 = L$	$q_5 P_1 = K$	$q_5 P_5 = H$

⑧

Combined DFA:



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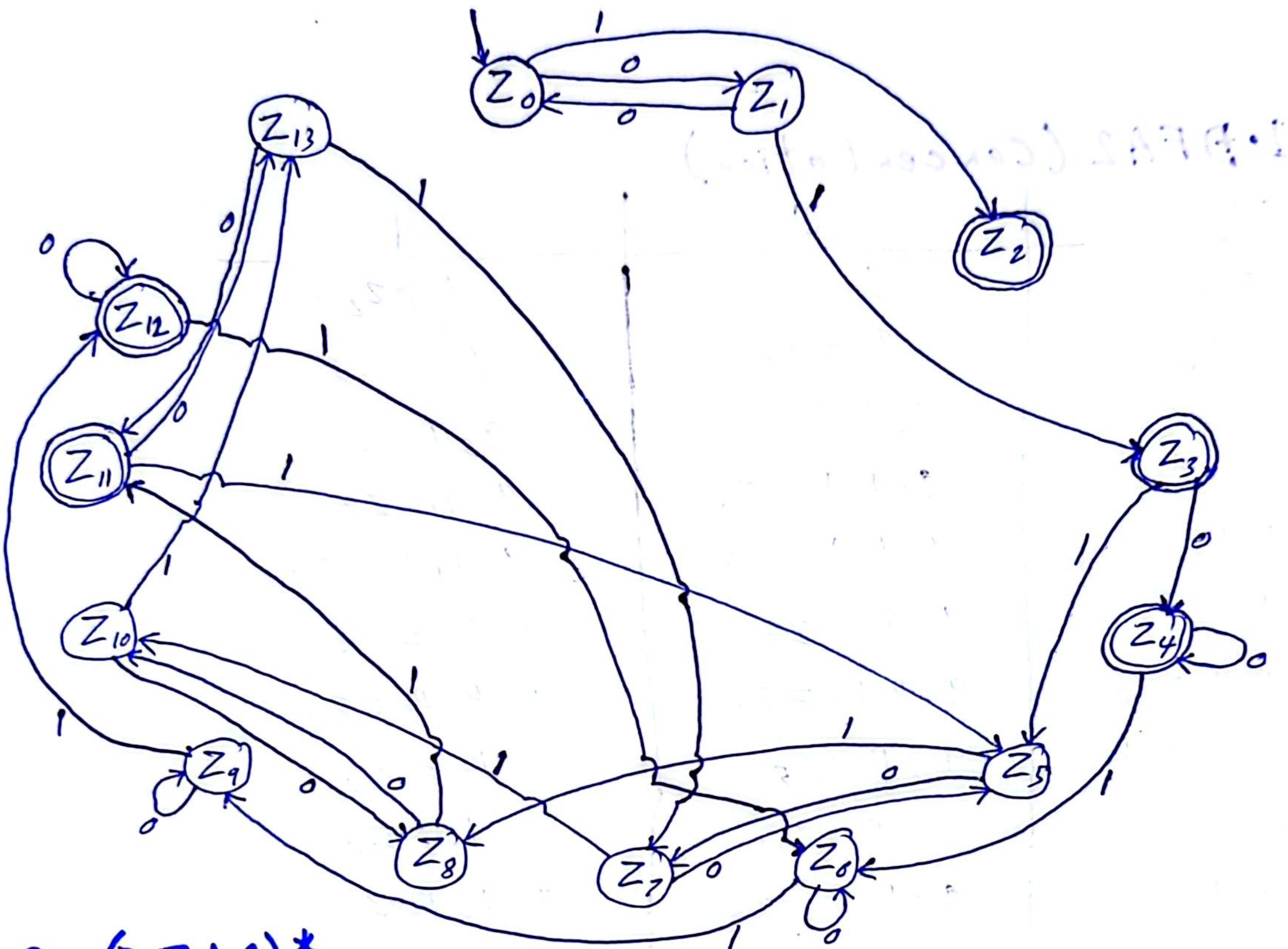
b. DFA1·DFA2 (Concentation)

	States	0	1
−	$q_0 = Z_0$	$q_3 = Z_1$	$q_1 P_0 = Z_2$
	$q_3 = Z_1$	$q_0 = Z_0$	$q_4 P_0 = Z_3$
+	$q_1 P_0 = Z_2$	$q_2 P_0 P_3 = Z_4$	$q_5 P_1 = Z_5$
+	$q_4 P_0 = Z_3$	$q_2 P_0 P_3 = Z_4$	$q_5 P_1 = Z_5$
+	$q_2 P_0 P_3 = Z_4$	$q_2 P_0 P_3 = Z_4$	$q_5 P_1 P_4 = Z_6$
	$q_5 P_1 = Z_5$	$q_5 P_4 = Z_7$	$q_5 P_2 = Z_8$
	$q_5 P_1 P_4 = Z_6$	$q_5 P_1 P_4 = Z_6$	$q_5 P_2 P_5 = Z_9$
	$q_5 P_4 = Z_7$	$q_5 P_1 = Z_5$	$q_5 P_5 = Z_{10}$
	$q_5 P_2 = Z_8$	$q_5 P_5 = Z_{10}$	$q_5 P_0 = Z_{11}$
	$q_5 P_2 P_5 = Z_9$	$q_5 P_5 P_2 = Z_9$	$q_5 P_0 P_3 = Z_{12}$
	$q_5 P_5 = Z_{10}$	$q_5 P_2 = Z_8$	$q_5 P_3 = Z_{13}$
+	$q_5 P_0 = Z_{11}$	$q_5 P_3 = Z_{13}$	$q_5 P_1 = Z_5$
+	$q_5 P_0 P_3 = Z_{12}$	$q_5 P_3 P_0 = Z_{12}$	$q_5 P_1 P_4 = Z_6$
	$q_5 P_3 = Z_{13}$	$q_5 P_0 = Z_{11}$	$q_5 P_4 = Z_7$

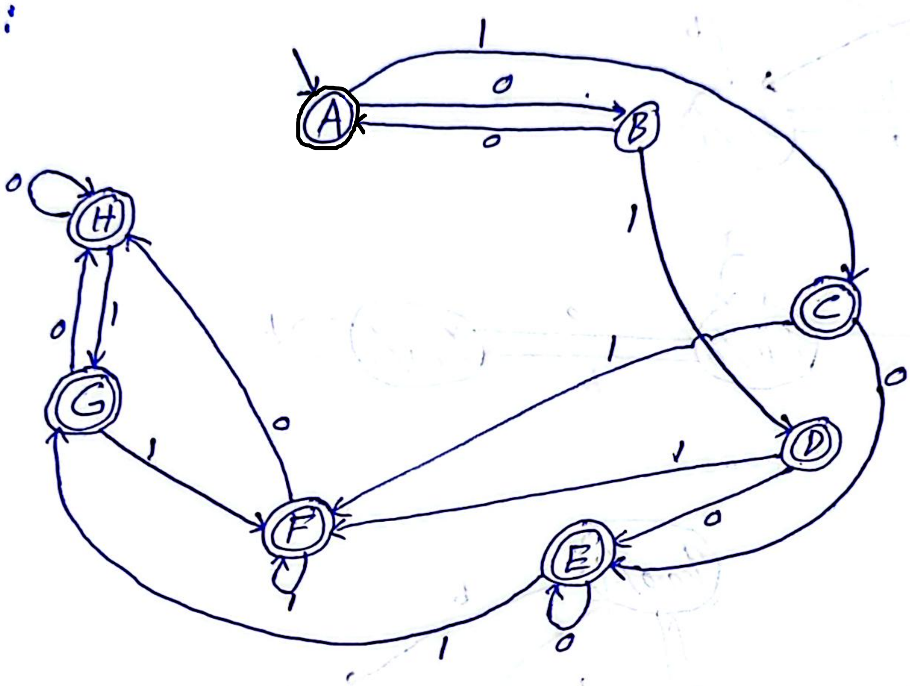
DFA1 · DFA2:



c. (DFA1)*

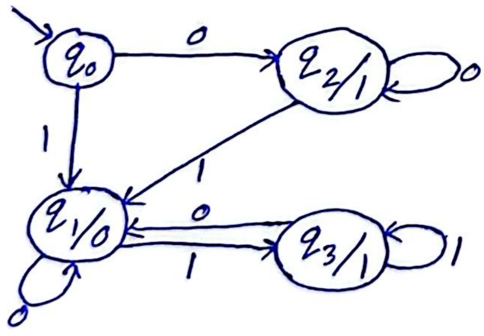
States	0	1
−+ $q_0 = A$	$q_3 = B$	$q_1 q_0 = C$
$q_3 = B$	$q_0 = A$	$q_4 q_0 = D$
+ $q_0 q_1 = C$	$q_3 q_2 q_0 = E$	$q_1 q_0 q_5 = F$
+ $q_0 q_4 = D$	$q_3 q_2 q_0 = E$	$q_1 q_0 q_5 = F$
+ $q_0 q_2 q_3 = E$	$q_3 q_2 q_0 = E$	$q_1 q_0 q_5 q_4 = G$
+ $q_0 q_1 q_5 = F$	$q_0 q_3 q_2 q_5 = H$	$q_1 q_0 q_5 = F$
+ $q_0 q_1 q_4 q_5 = G$	$q_3 q_2 q_0 q_5 = H$	$q_0 q_1 q_5 = F$
+ $q_0 q_2 q_3 q_5 = H$	$q_0 q_2 q_3 q_5 = H$	$q_0 q_1 q_4 q_5 = G$

(DFA 1)*:

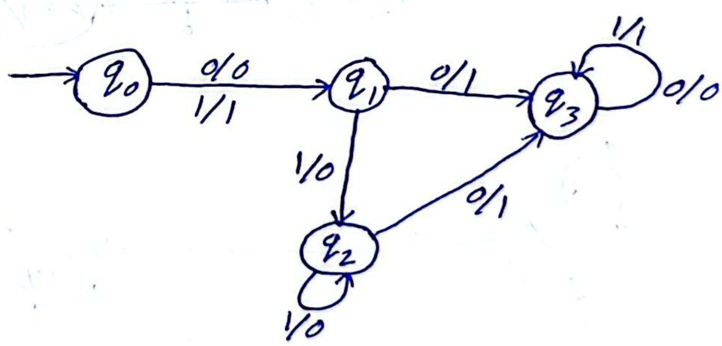


Q3:

B.



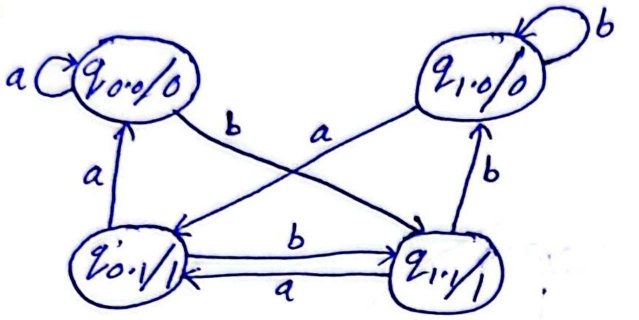
C.



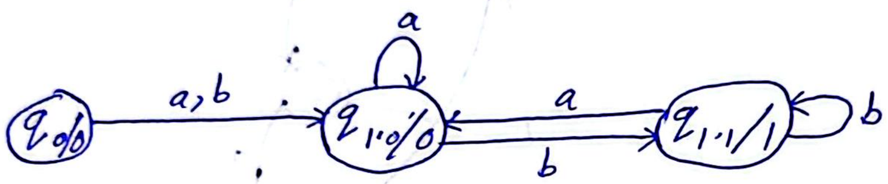
(12)

D. Mealy to Moore:

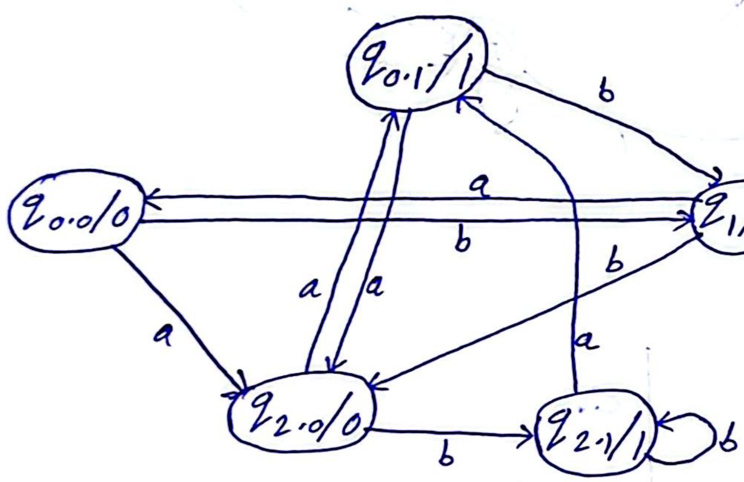
a.



b.



c.



d.